SYSTEMS MANAGEMENT



Prescribing Manageability for IT and OT

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Agenda

- Managing IT and OT aspects of the datacenter
- Profiles in 2026
  - Baseline Profile v1.1, Ethernet Profile v1.0, Service Baseline Profile v1.0
- Tools for creating Profiles
  - AI (examples from the HWLC and Neocloud workstreams)
  - Profile Creator tool

Managing the datacenter

- In the IT Infrastructure and OT Infrastructure quadrants, the orchestrator collects telemetry and performs autonomous action within their domain. The orchestrator may use telemetry from the other quadrant, but it can initiate action only in its quadrant.
- The orchestrator in the dynamic datacenter quadrant collects telemetry and perform action is both the IT and OT infrastructures. Hence it can pursue the optimal path to fulfill a policy or achieve a goal.

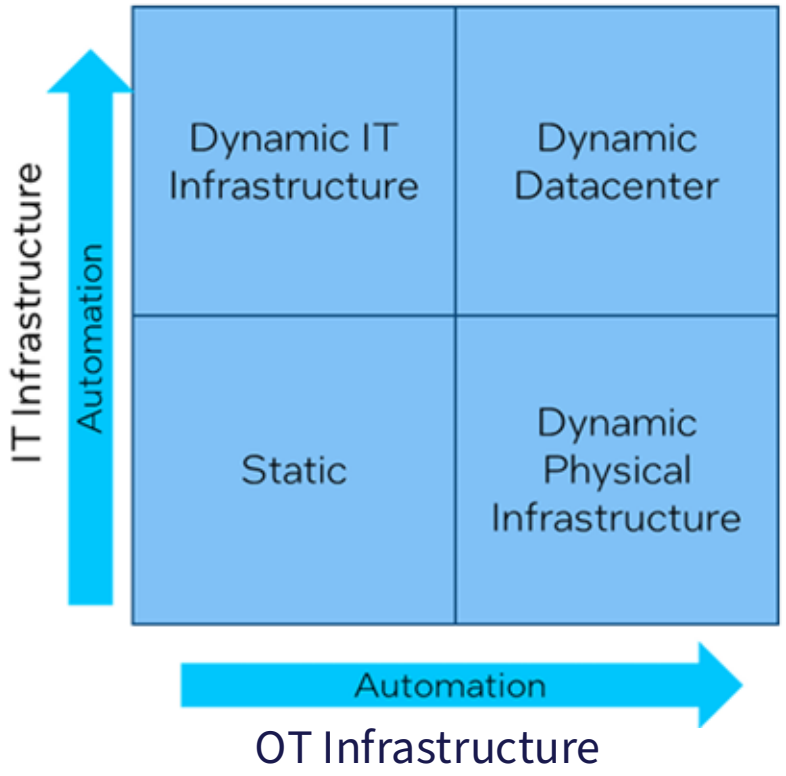
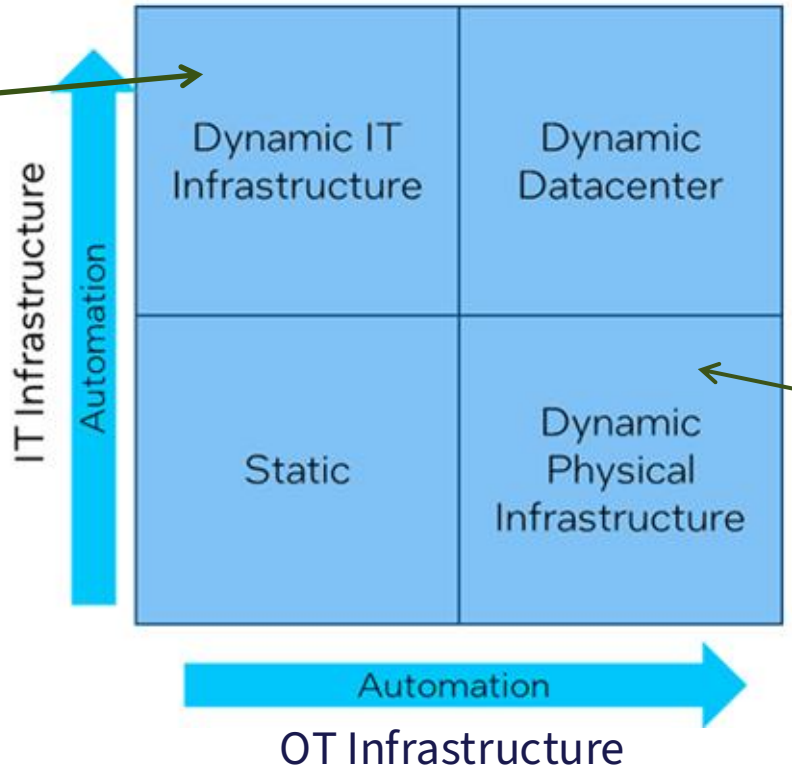


Diagram from “OCP Open Systems for AI” whitepaper

OCP Profiles for managing IT/OT

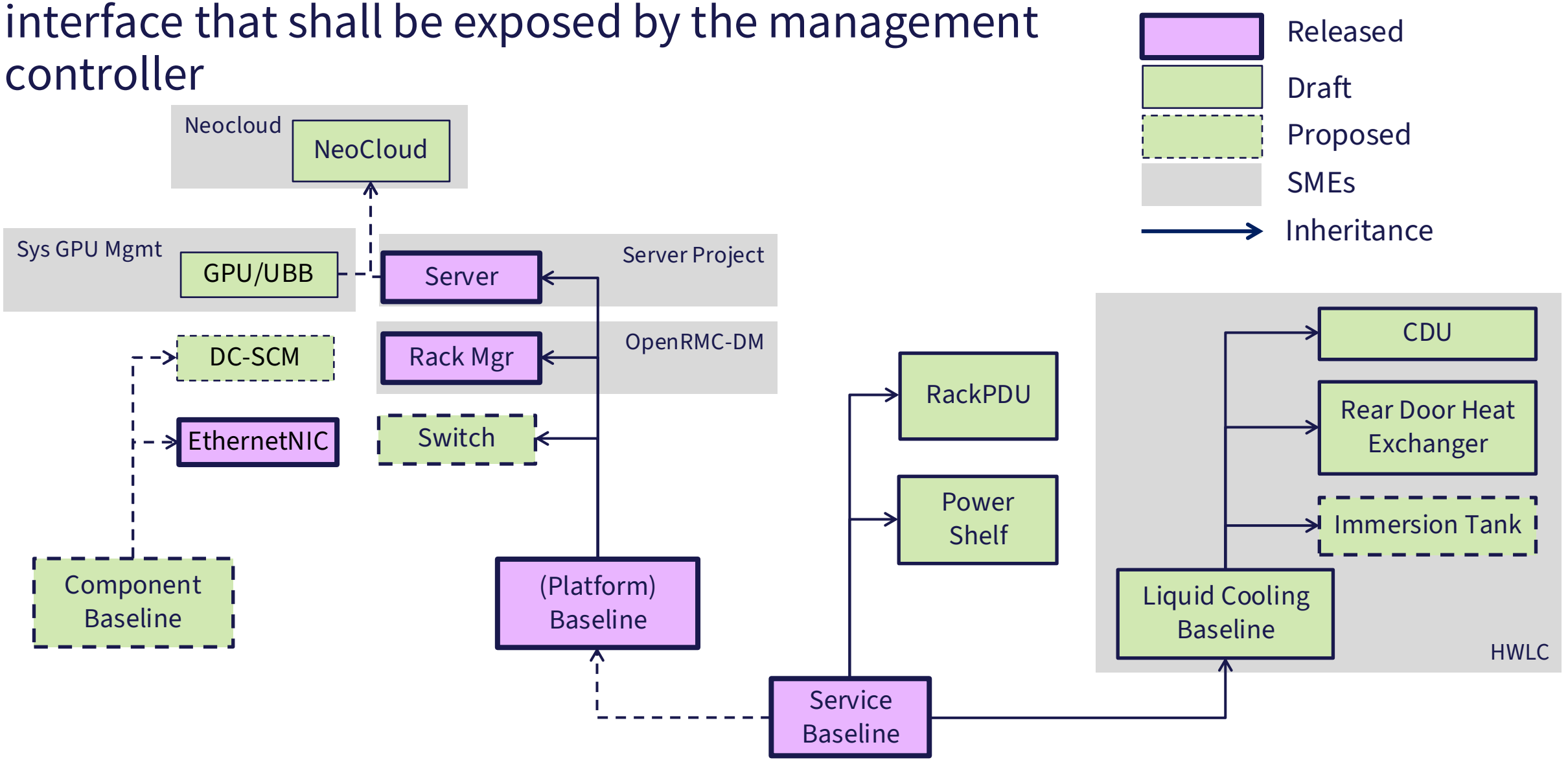
Managing IT Infrastructure
- HW Baseline
- Server
- Ethernet NIC
- Rack Manager
- GPU
- Ethernet Switch



Managing OT Infrastructure
- CDU
- Rear Door Heat Exchanger
- Power Shelf
- Rack PDU

DC Telemetry workstream (DCF Project) - https://www.opencompute.org/community/data-center-telemetry
- IT/OT Telemetry Integration
- Multi-tenant data sharing - while maintaining security and confidentiality.
- Business purpose for telemetry integration to tenants / customers

A OCP profile *prescribes* the Redfish manageability interface that shall be exposed by the management controller



https://github.com/opencomputeproject/HWMgmt-OCP-Profiles

In 2026, the Profile WS contributed three profiles

- **Ethernet NIC Profile v1.0**
  - Basic manageability of an Ethernet NIC, referenced by the Ultra Ethernet NIC
  - Verified via the mockup[1] and physical systems



e.g. Intel Ethernet E835 Network Adapters

- **Baseline Profile v1.1**
  - The Power and Thermal resources where deprecated (2024) in favor of PowerSubystem and ThermalSubsystem resources
  - Baseline v1.1 allows either Power/Thermal or PowerSubsystem/ThermalSubsystem resources (backward compatible)
- **Service Baseline Profile v1.0**
  - The current baseline profile only prescribes some Redfish services
  - The service baseline profile prescribes a common set of Redfish services that can be referenced (e.g. baseline profile v2.0)1

[1]Redfish Mockup Server

Service Baseline Profile

- Prescribes common management tasks expected by any Redfish service
  - The “housekeeping” functions
  - Not tied to a product or specific architecture
- Allows SMEs in other projects to focus on manageability requirements specific to their domain
  - Example: A GPU profile need not re-invent Redfish requirements to manage Redfish user accounts

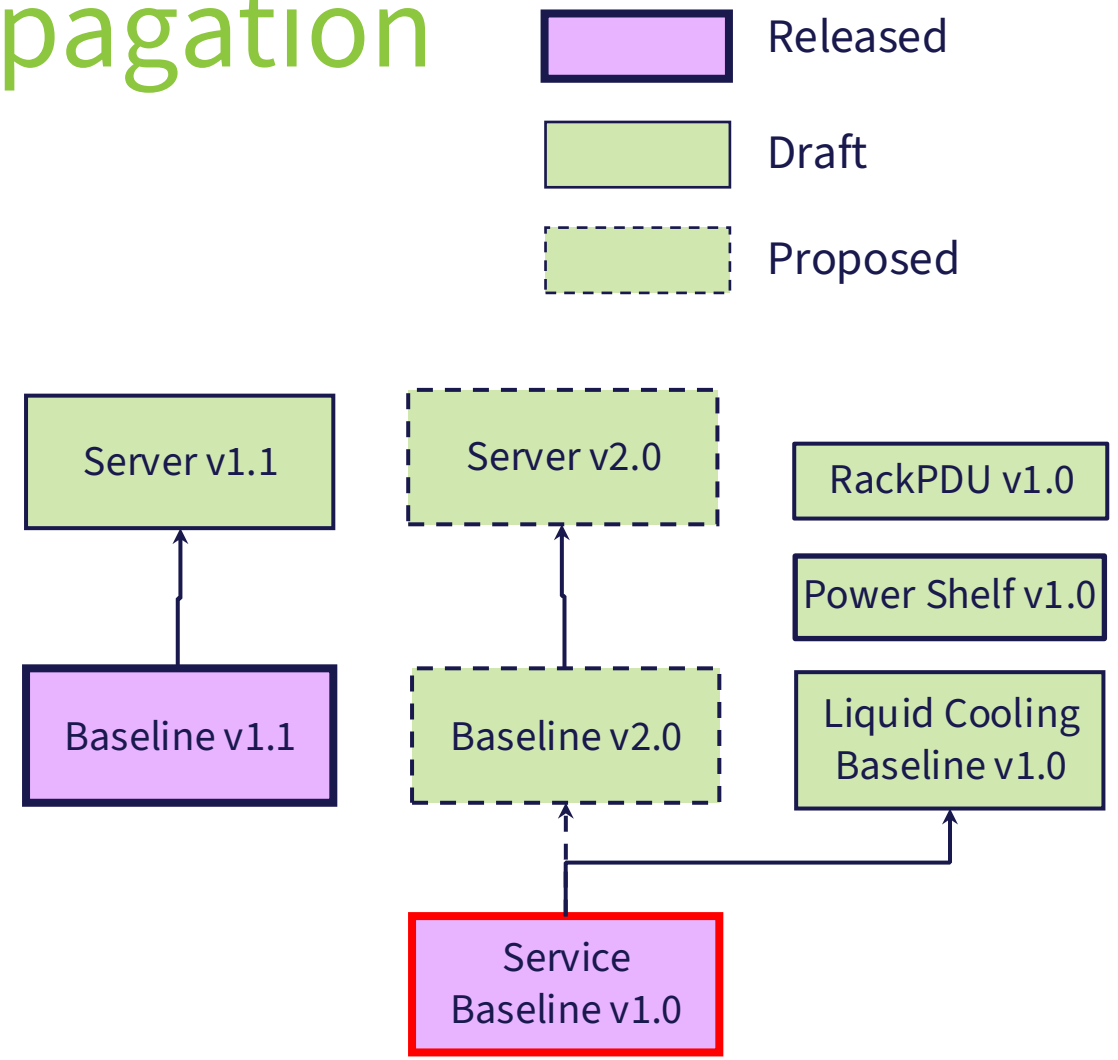
Service Baseline Profile 1.0 Use Cases

Management Task	Requirement
Get/create/remove users	Mandatory
Get/replace certificate	Mandatory
Generate CSR	Mandatory
Get/create/remove event subscriptions	Mandatory
Send events to subscribers	Mandatory
Get service IP settings	Mandatory
Set service static IP address	If Implemented
Set service to DHCP	Recommended
Get/install/uninstall license	Recommended
Get/clear event log	Mandatory

Management Task	Requirement
Get software/firmware inventory	Mandatory
Software/firmware update	Mandatory
Get service network protocols	Mandatory
Set service network protocols	Recommended
Get service conditions	Recommended
Get/create/remove registered client	Recommended
Get/create/delete session	Mandatory
Change session timeout	Recommended
Get tasks	If Implemented

Service Baseline Profile Propagation

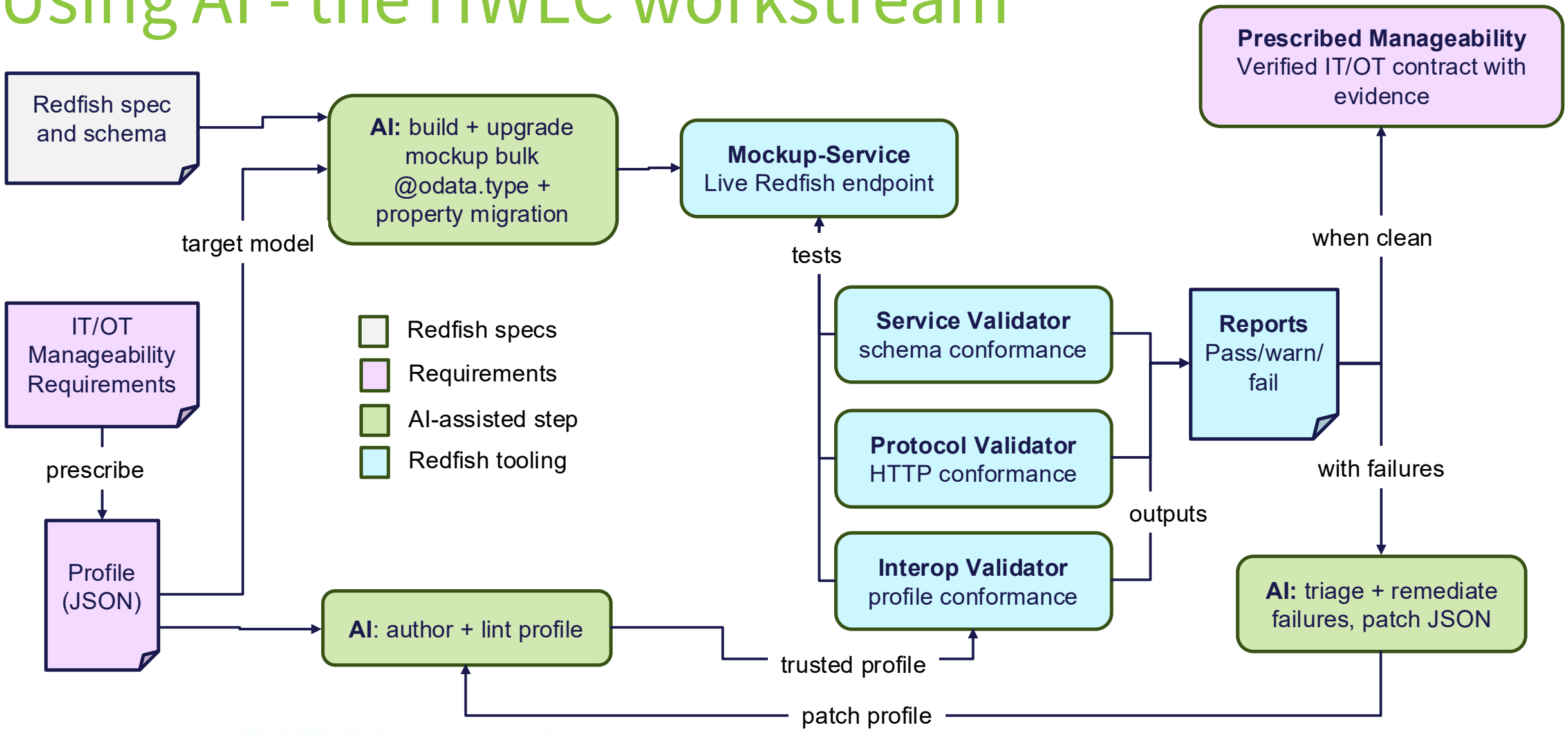
- The Service Baseline profile is referenced by the liquid cooling baseline, power shelf, and rack PDU profiles
- The Service Baseline profile will be referenced by the Baseline 2.0 profile



Using AI – Neocloud workstream

- The “Scaling AI Clusters at Neoclouds” workstream resides in Future Tech Init.
- Their initial polling highlighted manageability as a priority area and where OCP had impactful work.
  - Survey results
- Used AI (which one?) to compare the ClusterMax requirements with the requirements already prescribed in the OCP baseline, server, GPU/UBB profiles
  - Produced a report and a neocloud profile
  - The Report includes a prioritized list of major gaps, suggested structure of deliverables, and features of the Redfish Specification v1.16 that a profile should require
  - Lots of work to verify the statements in the report
- Concern – Lack of interoperability of GPU Profile

Using AI - the HWLC workstream



Redfish Profile Creator

Schema	Use Case	Property	Read Rqmt	Write Rqmt	Conditional	Purpose
Chassis	SampleCase	Id	Mandatory	<blank>	False	Identifies the Object

Handles the syntactical requirements of the profile syntax

python3 profile-creator.py <csv_file>

```
"Resources": {
  "Chassis": {
    "UseCases": [
      {
        "UseCaseTitle": "SampleCase",
        "UseCaseType": "Normal",
        "PropertyRequirements": {
          "Id": {
            "ReadRequirement": "Mandatory",
            "WriteRequirement": "None",
            "Purpose": "Identifies this Object"
          }
        }
      }
 ...
```

https://github.com/DMTF/Redfish-Tools/redfish-profile-creator

Samples folder contains CSVs and profiles to demonstrate tool

File	Description
SampleProfile.json	Initial profile to override with other profiles
ExampleOverwrite.csv	Input CSV file to demonstrate overwriting behavior of items found in SampleProfile.json
ExampleProfileNoCase.json	Input CSV file to create an example output, with blank usecases
ExampleProfileCase.json	Input CSV file that creates an example output, with usecases

Call to Action

- Follow progress of the DC (IT/OT) Telemetry workstream
  - Co-lead (Scott Sharp)



- Tests draft profiles and identify issues before the release.
  - The testing of Server Profile v1.1 draft uncovered issues with the profile and the Interop Validator
- Start workstreams with your projects to prescribe manageability
- Contacts for more tooling info
  - Lead of the HWLC workstream (Mick Jones)



  - Co-lead Neocloud workstream (Jonmichael Hands)



  - Redfish Profile Creator (Jeff Autor)



Thank You!

